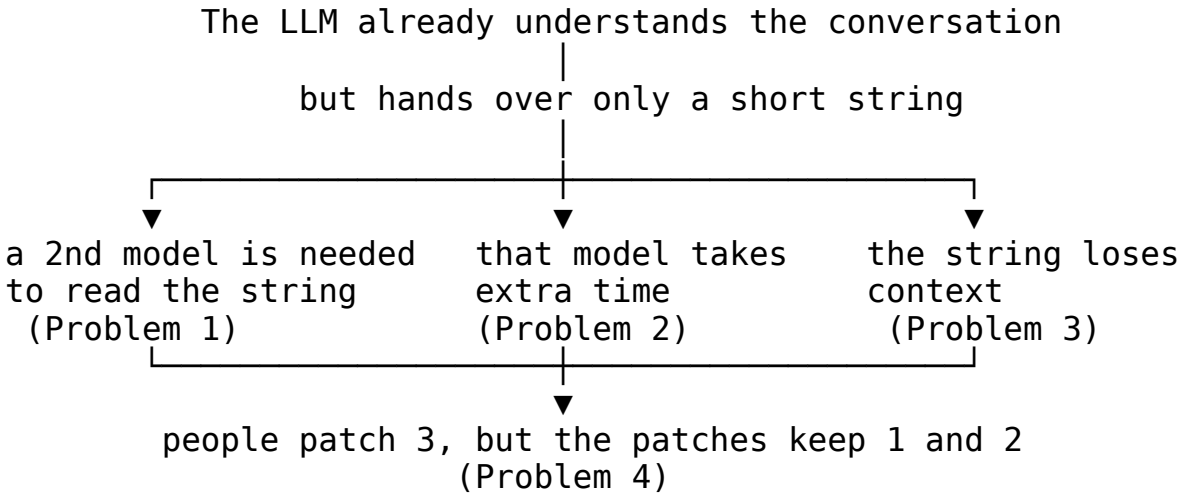
05 · The Problems at a Glance

The paper's main complaint is a single word: **redundancy**. The LLM understands the situation, writes a summary of that understanding as text, and then a second model re-reads the summary to rebuild an understanding the first model already had.

That redundancy shows up as four problems.

#	Problem	One-line description	Kind of cost
1	**Two models**	You must deploy, pay for, and keep in sync two large networks.	Money, engineering
2	**Latency**	The second model adds a full forward pass to every search.	Time
3	**Information bottleneck**	The embedder only sees the short query text, not the conversation.	Search quality
4	**Patches don't remove the split**	Known fixes improve quality but keep, or even add to, the two-model setup.	All of the above

How they connect



Problems 1 and 2 are about **cost**. Problem 3 is about **quality**. Problem 4 explains why the field hasn't simply fixed this already.

A fair warning before you read on

The paper argues all four, but its experiments measure them unevenly:

- Problem 2 (latency) is **measured** directly: 43.5 ms vs 2.0 ms.
- Problem 1 (two models) is removed **at inference time only**; the teacher is still needed for training.
- Problem 3 (bottleneck) is the **motivation**, but the experiments don't show the new method beating the baseline on quality. It comes out slightly worse. File 19 discusses this.

Keep asking "was this measured?" as you read.